Intro to LaTeX

HSE 103: Fall 2026

Lecture Objectives

At the end of this lecture you should be able to:

1. Create a basic document in LaTeX
2. Typeset mathematical equations
3. Apply basic formatting
4. Generate references

Resources

LaTeX Editors

Overleaf: https://www.overleaf.com/edu/dartmouth
As as a Dartmouth student, you have free access to Overleaf.

Open Source Reference Managers

Mendeley: https://www.mendeley.com/

Zotero: https://www.zotero.org/

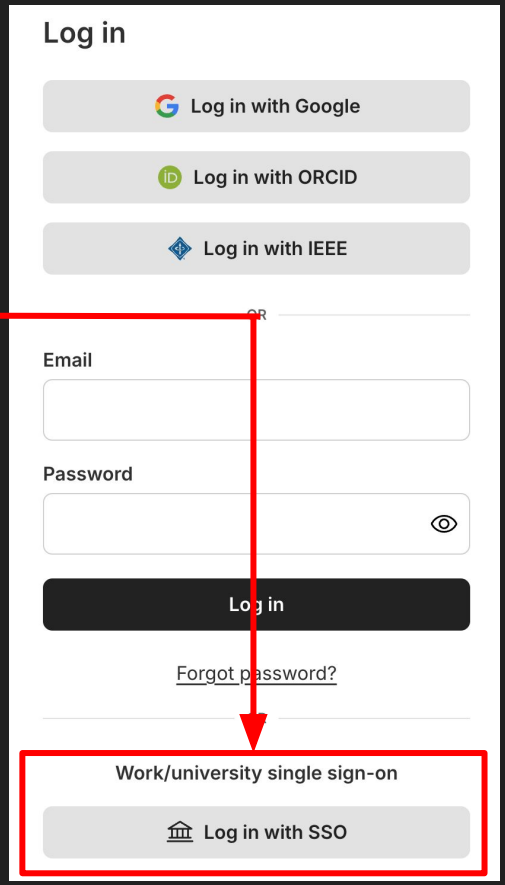
The Basics: Why use LaTeX?

- Create clean, formatted scientific documents
- Provide the same entered base text and easily apply any formatting afterwards (i.e. focus on content first, don’t worry about formatting as you write)
- Easy typesetting for mathematical equations and references, consistently formatted across all documents

Accessing Overleaf

Log in to Overleaf with your Dartmouth credentials using “Log in through your institution”.

As long as you log in through Dartmouth, you should see that you have access to a premium overleaf subscription.



Starting a New Document

LaTeX files use the extension .tex

At a minimum, every new .tex file you start needs to contain the following lines:

```
\documentclass{article}
\begin{document}
[Document Content]
\end{document}
```

Unlike in R where we use # to indicate comments, in LaTeX we use %

Building Our Document

We can add additional features to our document using features like these:

```
\documentclass{article}

\title{your title here}
\author{your name here}
\date{your date here}

\begin{document}

\maketitle % inserts text for your title as defined above

\tableofcontents %automatically generates based on sections, subsections, etc defined in your document

\section{your section name here}
\subsection{your subsection name here}

You can also add text as you want it to appear throughout just by typing.

\end{document}
```

Applying Formatting

The *geometry* package can be used to define a basic layout for your document like this:

```
\usepackage{geometry}

\geometry{total = {8.5in,11in}, margin=1in}
```

Applying Formatting

You can set a base font size in the “document class” line like this:

\documentclass[size]{class}

“article” accepts sizes: 10pt, 11pt, 12pt

“extarticle” accepts sizes: 8pt, 9pt, 10pt, 11pt, 12pt, 14pt, 17pt, 20pt

Then, you can change the relative size of text within the document using commands to the right.

Command	Example text	Font size
\tiny	Some text	5pt
\scriptsize	Some text	7pt
\footnotesize	Some text	8pt
\small	Some text	9pt
\normalsize	Some text	10pt
\large	Some text	12pt
\Large	Some text	14.4pt
\LARGE	Some text	17.28pt
\huge	Some text	20.74pt
\Huge	Some text	24.88pt

Relative Sizes Based on 10pt Base Font

Change font: https://www.overleaf.com/learn/latex/Font_typefaces

Typesetting Equations

Whenever we're typing an equation we use the following syntax to denote that the text inside reflects a mathematical equation.

On a New Line:
\[your equation here \]

In Line with Text:
$ your equation here $

\(your equation here \)

\begin{math} your equation here \end{math}

https://www.overleaf.com/learn/latex/Mathematical_expressions

Typesetting Equations

Mathematical Symbols
https://www.overleaf.com/learn/latex/List_of_Greek_letters_and_math_symbols#Binary_Operation/Relation_Symbols

Brackets and Parentheses
https://www.overleaf.com/learn/latex/Brackets_and_Parentheses

Integrals, Subscripts, and Superscripts
https://www.overleaf.com/learn/latex/Subscripts_and_superscripts

Typesetting Equations

If we want to spread something over multiple lines centered on the =, we can use equation and split from the amsmath package:

```
\usepackage{amsmath}

\begin{equation}

    \begin{split}

        text to left &= text to right \\

        &= text to right \\

    \end{split}

\end{equation)
```

Using Typeset Equations in R Markdown

To drop an equation you've already typeset into markdown, simply enter it with the following syntax in a text chunk (not a code block) of your markdown file to start it on a new line:

$$ your equation here $$

You will see the formatted equation appear immediately in your markdown window (no need to run any code) and in your final knitted document.

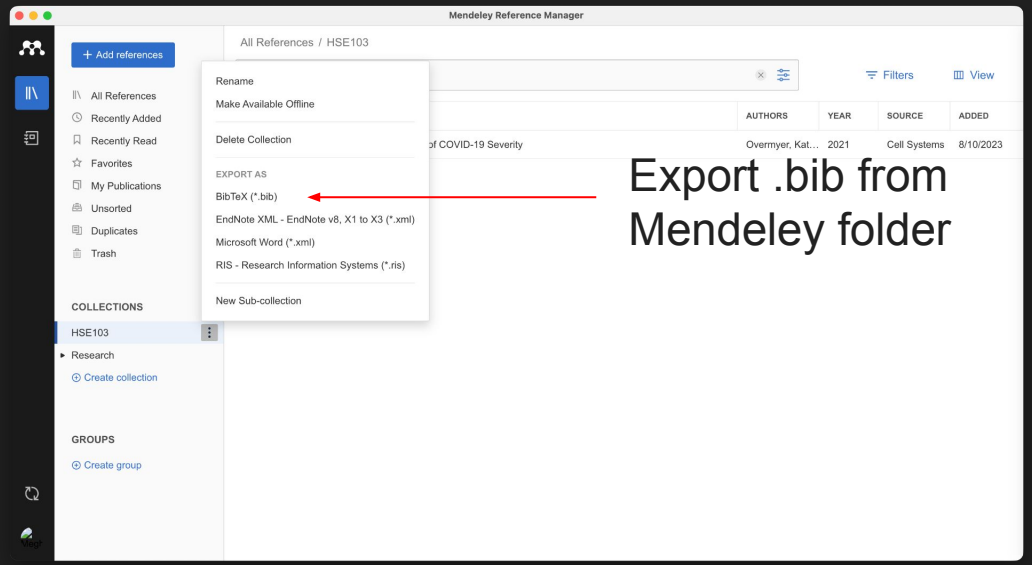
Building Tables in LaTeX

Tutorial: https://www.overleaf.com/learn/latex/Tables

Easy Table Generator: https://www.tablesgenerator.com/

Adding in References

To add formatted references to a LaTeX, all you need is a .bib file which you can easily export from most citation managers. Add this file to your folder.



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At the end of your document, add this:

\bibliographystyle{format you want to use}
\bibliography{file name (no .bib extension)}

Note, the bibliography will be blank until you cite specific articles. Which you can do with the following command in line:
\cite{citation name}

Adding Images

To add an image to your LaTeX document, first add it to your folder in Overleaf. If you have a lot of images, it can help to make separate folders to sort them.

Add this to the top of your document:

```
\usepackage{graphicx}
\graphicspath{ {./images/} }
```

And call it inline like this

```
\includegraphics{image path}
```

Adding Images

If you want to specify more things about your figure, you can format it like this:

```
\begin{figure}[h] % h means roughly here, H means exactly here
    \centering
    \includegraphics[width=\textwidth]{image path}
    \caption{Your caption here}
    \label{fig:nickname}
\end{figure}
```

In Class Activity

Typeset the following proof in LaTeX:

Law of Total Probability

Theorem: If $B_1, B_2, B_3, \ldots$ is a partition of the sample space S, then for any event A we have

$$P(A) = \sum_i P(A \cap B_i) = \sum_i P(A|B_i)P(B_i)$$

Proof:

Since $B_1, B_2, B_3, \ldots$ is a partition of the sample space S, we can write

$$\begin{aligned} S &= \bigcup_i B_i \\ A &= A \cap S \\ &\Rightarrow A \cap \left(\bigcup_i B_i\right) \\ &= \bigcup_i (A \cap B_i) \end{aligned}$$

Note that because B_i's are disjoint, $A \cap B_i$ are disjoint. Hence,

$$\begin{aligned} P(A) &= P\left(\bigcup_i (A \cap B_i)\right) \\ &= \sum_i P(A \cap B_i) \\ &= \sum_i P(A|B_i)P(B_i) \end{aligned}$$

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